Cluster data

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Introduction

In this document, I estimate the abundance of Black-eared Miners from cluster data.

Load packages and data

```
library(Distance)
library(dplyr)
library(tidyr)

# read data
raw_data <- read.csv("Data/All_sightings.csv", header = TRUE)

# randomly select one of the transects that were walked more than once
# to standardise effort
sample_dupl <- raw_data %>%
  filter(Ref %in% c("RLB034", "RLB147", "RLB224", "RLB290", "RLB417", "RLB586")) %>%
  group_by(Ref) %>%
  slice_sample(n=1)

# remove duplicated transects and add back the sampled ones
ds_data0 <- rbind(raw_data[!raw_data$Ref %in%
                             c("RLB034", "RLB147", "RLB224", "RLB290", "RLB417", "RLB586")
                  sample_dupl)

ds_data <- data.frame(Sample.Label = ds_data0$Ref,
                      Region.Label = case_when(ds_data0$Area == 300 ~ "BURNT",
                                               ds_data0$Area == 600 ~ "OLD"),
                      distance = ds_data0$Distance,
```

```
                        Effort = rep(1000, nrow(ds_data0)),
                        size = ds_data0$No..of.miners,
                        Area = ds_data0$Area)

nrow(ds_data[!is.na(ds_data$distance),]) # how many datapoints
```

```
[1] 131
```

Explore the data

```
# plot
par(mfrow = c(2,2))
hist(ds_data$distance, breaks = seq(0,400,10), main="", xlab="Distance (m)")
hist(ds_data$distance, breaks = seq(0,400,20), main="", xlab="Distance (m)")
hist(ds_data$distance, breaks = seq(0,400,25), main="", xlab="Distance (m)")
hist(ds_data$distance, breaks = seq(0,400,50), main="", xlab="Distance (m)")
```

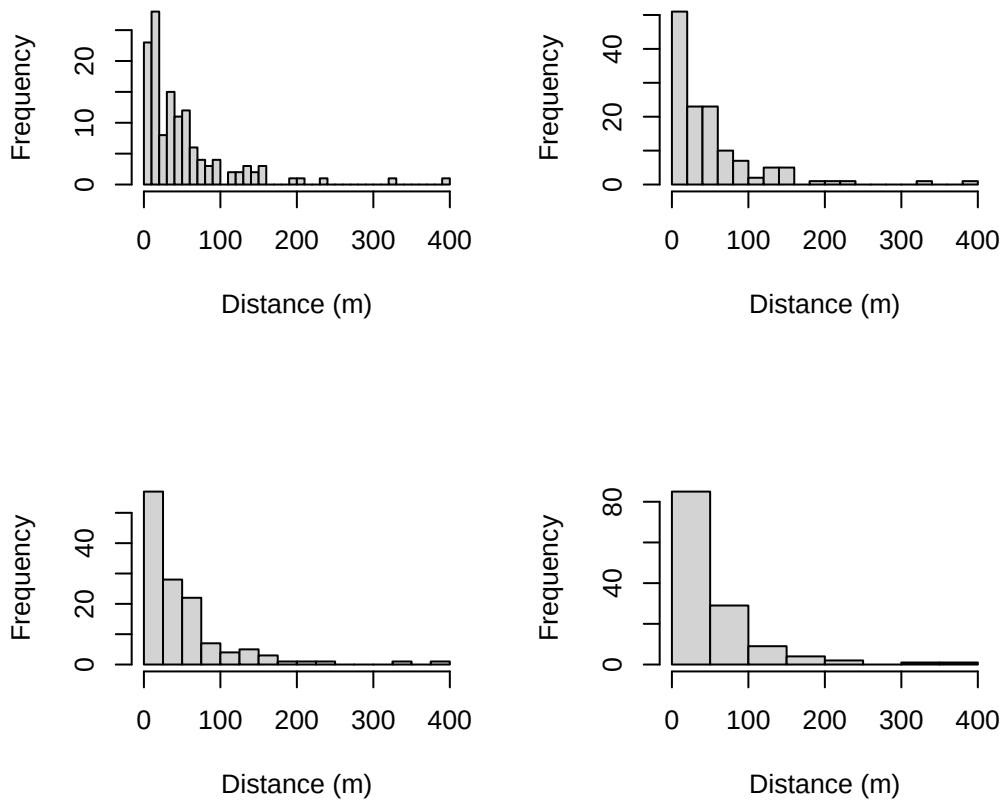


Figure 1: Frequency distribution of distances to clusters.

We can see spiked data close to the line. Thomas et al. (2010) states: "For such data, different models will give very different estimates of density, so it is important to understand what has caused the spike, and to modify field procedures accordingly." The following can be considered in the field sampling procedure in future surveys:

> **Note**
>
> "Spiked data may arise even when there has been no failure of an assumption. For example, in surveys of breeding songbirds, singing males may be much more detectable than (non-singing and cryptic) females. In that case, the spike arises because females are only detectable close to the line. The simplest solution in this case is to additionally record whether the bird was singing. An analysis can then be conducted for singing birds,

> allowing estimation of the number of territories. If females are certain to be detected when on the line, then a separate analysis of females could be conducted, if sample size is adequate, or sex could be included as a covariate in an MCDS analysis. Similar issues apply for point transect surveys." **(Thomas et al. 2010)**

For now, I try to deal with the spike. A distance bin interval of 25m at a truncation distance of 125m appears to fit nicely to a half-normal key function.

```
mybreaks <- seq(0,400,25)

# plot
hist(ds_data$distance, breaks = mybreaks, main="", xlab="Distance (m)")
abline(v=125, col="red")
```

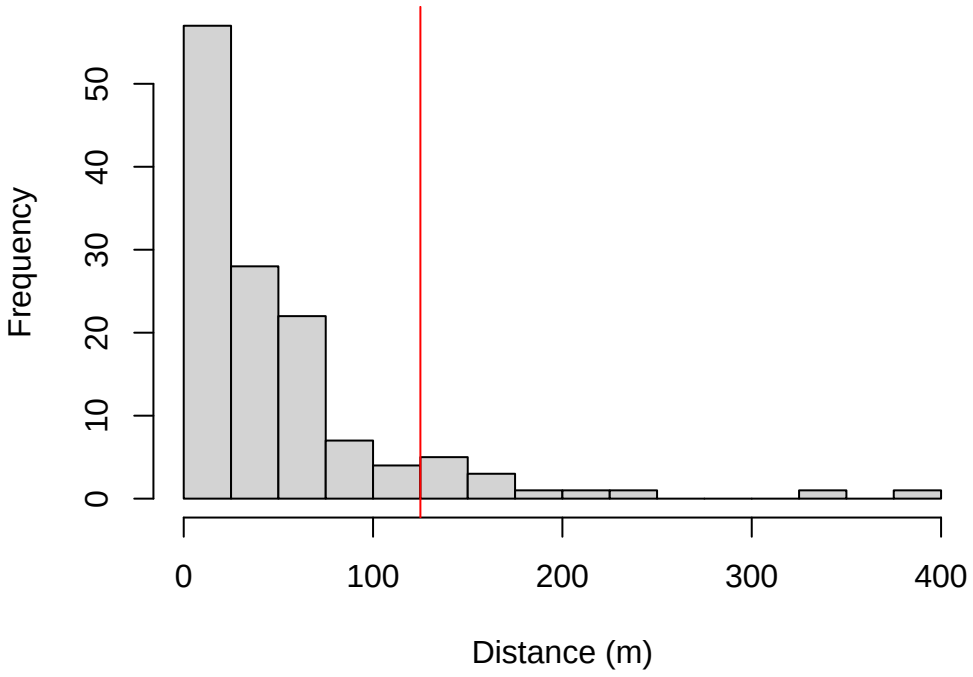


Figure 2: Frequency distribution of distances to clusters.

More information on picking the right truncation distance for detection model fitting for ref-

erence:

> **Note**
>
> "We truncate because otherwise extra adjustment terms may be needed to fit a long tail to the detection function. This reduces precision for little gain, as data a long way from the line or point contribute little to the abundance estimate (Buckland et al. 2001:103–108, 151–153). We typically truncate around 5% of distances for line transect sampling, and more for point transect sampling (for which a higher proportion of detections corresponds to the tail of the detection function, Buckland et al. 2001:151)." **(Thomas et al. 2010)**

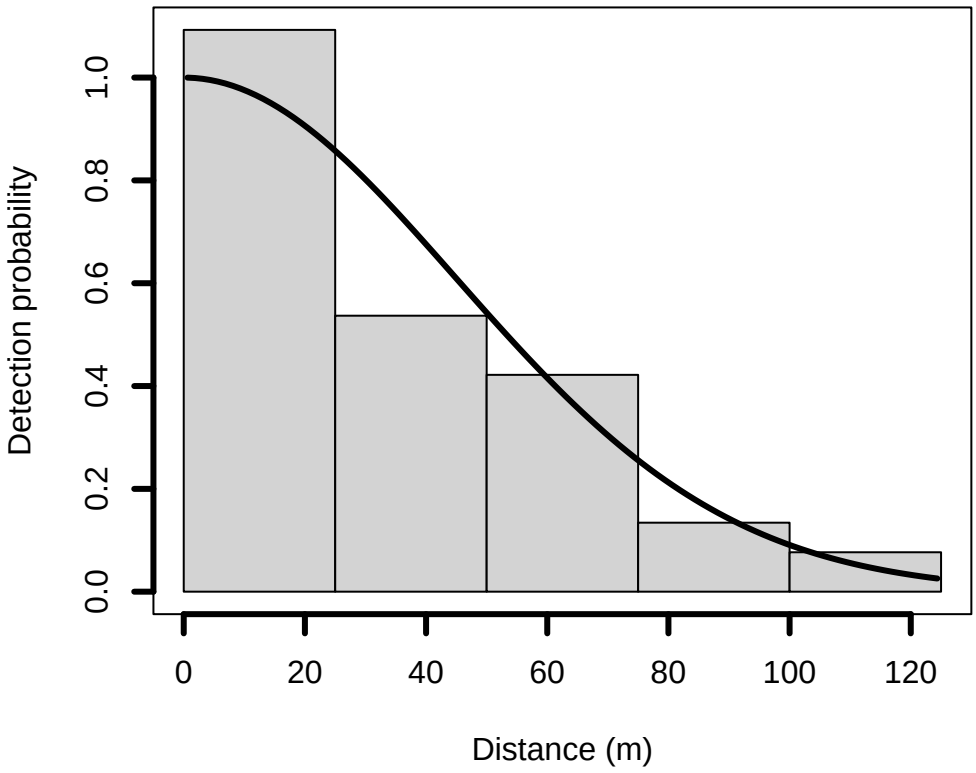
Distance analysis

I first fit a model with habitat type(?) treated as a stratification factor (i.e., under `Region.Label`). This means that detection functions fitted for both habitats are the same key function, but the scale parameter of the key function differs between years. I've done this to allow for the detection model to vary between habitat types, in case there is a difference.

```
conversion <- convert_units("meter", "meter", "square kilometer")
mybreaks <- seq(0,125,25)

# half-normal
hn_strat <- ds(ds_data,
                truncation = list(left=0, right=125),
               convert_units = conversion, cutpoints = mybreaks,
               key = "hn", adjustment = "cos", #order = 2,
               formula=~as.factor(Region.Label))

# check fit
plot(hn_strat, main="", xlab="Distance (m)",
     showpoints=FALSE, lwd=3, xlim=c(0, 125),pdf=F)
```



Examining the `summary()` output, we see the region-specific estimates being provided:

```
summary(hn_strat)
```

```
Summary for distance analysis
Number of observations :  118
Distance range         :  0  -  125

Model : Half-normal key function
AIC   : 311.5634

Detection function parameters
Scale coefficient(s):
```

```
                            estimate        se
(Intercept)                 3.731532 0.1201580
as.factor(Region.Label)OLD 0.146937 0.1507984

                      Estimate         SE         CV
Average p             0.452429  0.0316915 0.07004746
N in covered region 260.814428 25.5081587 0.09780195

Summary for clusters

Summary statistics:
  Region Area CoveredArea Effort   n   k           ER        se.ER     cv.ER
1  BURNT  300       80.00 320000  48 320 0.0001500000 2.480330e-05 0.1653553
2    OLD  600       74.75 299000  70 299 0.0002341137 3.171316e-05 0.1354605
3  Total  900      154.75 619000 118 619 0.0002060758 2.270120e-05 0.1101595

Abundance:
  Label  Estimate        se        cv       lcl       ucl       df
1 BURNT  431.2551  87.33273 0.2025083  290.8044  639.5397 425.3517
2   OLD 1170.4059 186.80158 0.1596041  856.9172 1598.5791 413.9994
3 Total 1601.6610 206.20823 0.1287465 1245.0155 2060.4706 515.7737

Density:
  Label Estimate        se        cv       lcl      ucl       df
1 BURNT 1.437517 0.2911091 0.2025083 0.9693481 2.131799 425.3517
2   OLD 1.950676 0.3113360 0.1596041 1.4281953 2.664299 413.9994
3 Total 1.779623 0.2291203 0.1287465 1.3833506 2.289412 515.7737

Summary for individuals

Summary statistics:
  Region Area CoveredArea Effort   n   k           ER        se.ER     cv.ER
1  BURNT  300       80.00 320000 221 320 0.0006906250 0.0001275359 0.1846673
2    OLD  600       74.75 299000 331 299 0.0011070234 0.0001920561 0.1734887
3  Total  900      154.75 619000 552 619 0.0009682239 0.0001349105 0.1393381
  mean.size   se.mean
1  4.604167 0.4291460
2  4.728571 0.4745801
3  4.677966 0.3300317

Abundance:
  Label Estimate        se        cv      lcl       ucl       df
1 BURNT 1985.570  433.9697 0.2185617 1298.665  3035.801 434.1608
```

```
2   OLD 5534.348 1067.7457 0.1929307 3800.544  8059.110 398.4211
3 Total 7519.918 1152.5670 0.1532686 5574.308 10144.608 497.5481

Density:
  Label Estimate       se        cv      lcl      ucl       df
1 BURNT 6.618568 1.446566 0.2185617 4.328884 10.11934 434.1608
2   OLD 9.223913 1.779576 0.1929307 6.334241 13.43185 398.4211
3 Total 8.355465 1.280630 0.1532686 6.193676 11.27179 497.5481

Expected cluster size
  Region Expected.S se.Expected.S cv.Expected.S
1  BURNT   4.604167     0.3904879    0.08481185
2    OLD   4.728571     0.4716693    0.09974879
3  Total   4.695075     0.3606344    0.07681121
```

The summary also shows beta parameters associated with region-specific estimates of the shape parameter of the half-normal key function. The standard errors of the beta (for `OLD`) is larger than the estimate itself. This suggests there us not a strong 'region' signal on the shape of the detection functions. Perhaps a model with a pooled detection function would be equally suitable for these data. See reference here.

```
# fit pooled detection function
hn_pooled <- ds(ds_data,
                truncation = list(left=0, right=125),
               convert_units = conversion, cutpoints = mybreaks,
               key = "hn", adjustment = "cos")

gof_ds(hn_pooled)
```

```
Goodness of fit results for ddf object

Chi-square tests
          [0,25] (25,50] (50,75] (75,100] (100,125]   Total
Observed  57.000  28.000  22.000    7.000     4.000 118.000
Expected  49.428  36.879  20.529    8.524     2.640 118.000
Chisquare  1.160   2.138   0.105    0.273     0.701   4.376

P = 0.2236 with 3 degrees of freedom
```

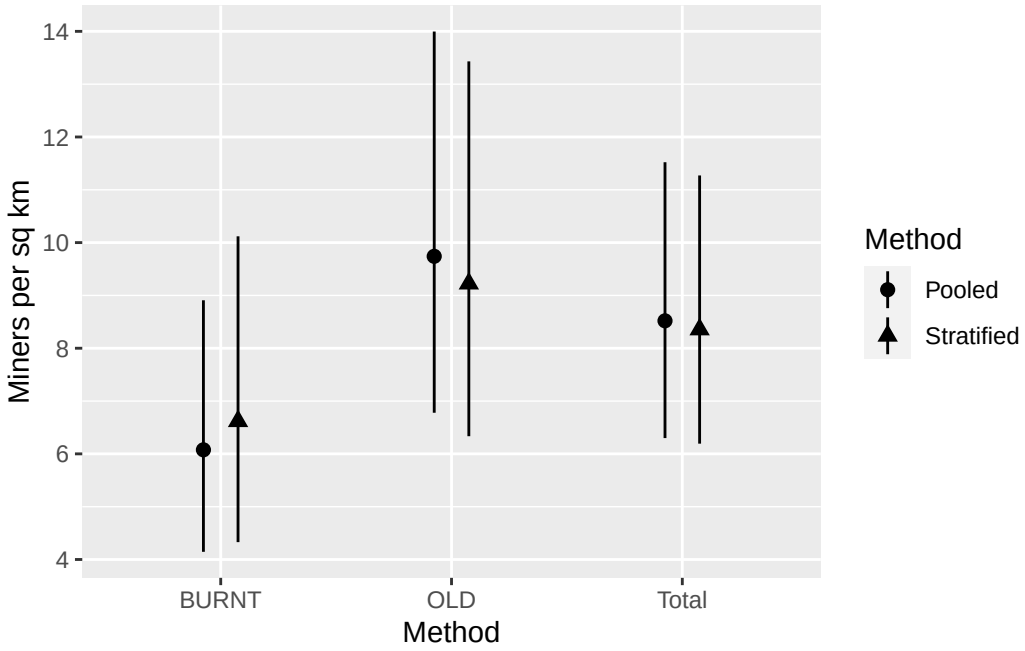
Having a look at the point and interval estimates of the two models bears out the minute difference between the models.

```r
dens_strat <- hn_strat$dht$individuals$D
dens_pooled <- hn_pooled$dht$individuals$D

compare_est <- rbind( data.frame(year = dens_strat$Label,
                                 estimate = dens_strat$Estimate,
                                 lcl = dens_strat$lcl,
                                 ucl = dens_strat$ucl,
                                 Method = "Stratified"),
                      data.frame(year = dens_pooled$Label,
                                 estimate = dens_pooled$Estimate,
                                 lcl = dens_pooled$lcl,
                                 ucl = dens_pooled$ucl,
                                 Method = "Pooled") )

library(ggplot2)

ggplot(compare_est, aes(x=year, y=estimate, shape = Method)) +
  geom_pointrange(aes(ymin=lcl, ymax=ucl), color="black",
                  position=position_dodge(width=0.3)) +
  labs(x="Method", y = "Miners per sq km")
```



We can go with the pooled model, which has a slightly smaller CV($\hat{D}$); i.e., higher precision.

Comparing the 2015 density estimates with that of the previous document ("1_Nest-vs-First.pdf"), we see a 19% increase in point estimates (8.52 vs 7.16) by using all the data (as opposed to removing clusters with <7 birds). There is also a marked improvement in precision (CV = 0.15 vs 0.25) when all observations are included; though note that I right-truncated the data at 125m to get a good detection model fit.

What's next

I believe this approach is a more appropriate than the original one that discards clusters with <7 birds as mentioned in my email correspondence. Furthermore, a good precision (CV = 0.15) now gives us flexibility to reduce survey effort.

I'll start work on exploring the reduction of survey effort via bootstrapping the data. Though, a word of warning: the distance analysis has to be tailored to the characteristics of each dataset/bootstrap sample (as I have demonstrated above) to obtain the best model fit. The variation in distributions of the bootstrap samples will be higher at low survey effort.